



Study of laminar single phase frictional factor and Nusselt number in In-line micro pin-fin heat sink for electronic cooling applications

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ABSTRACT

In the present study, hydrothermal characteristics of elliptical pin fin micro heat sink applied with constant heat flux from the bottom are analysed. Experiments are performed on a silicon elliptical pin fin micro sink using de-ionised water as the coolant. A numerical model is developed for single phase incompressible laminar flow and validated against in-house experimental results. The validated numerical model is used to study the effect of geometrical features on the hydrodynamic and thermal characteristics of micro pin fin heat sinks. The behaviour of local heat transfer coefficient is found to be similar after few initial rows of fins from the inlet of heat sink and is dependent on the size of recirculation zone formed behind the fins. A criterion for selecting length scale for defining Reynolds number, Nusselt number and frictional factor is proposed and correlations describing the behaviour of the Poiseuille number and Nusselt number with respect to the Reynolds number and Prandtl number are developed. The detailed understanding and empirical correlations will be useful in design of micro pin fin heat sink with elliptical fins.

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1. Introduction

There exists several effective cooling methods, such as single/multiphase liquid convection, thermoelectric, micro-refrigerators for high heat flux dissipation from electronic devices. The potential of single phase microchannel heat sink was first reported by Tuckerman and Pease [1]. High heat transfer coefficient can be obtained from microchannel based heat sinks due to the large surface area to volume ratio of microchannel. Thus, laminar single phase forced convection in microchannel is considered to be an effective way of cooling with applications in particle accelerators, electromagnets, nuclear reactors, power plants, process and production companies and electronic devices. Microchannels are extensively studied experimentally [2–3] and numerically [4–6]. For high heat flux dissipation with single phase, analysis of forced convection with different geometries of heat sink, viz. pin fin of different cross-sections [7] and metal foams [8] is reported in the literature. For high heat dissipating devices, micro pin fin heat sink offers very low thermal resistance as compared to rectangular channel heat sink [9], because of high heat transfer coefficient, however pumping power is high. Owing to the high heat dissipation demand,

micro-pin fin heat sink can be used more effectively as compared to the conventional microchannel heat sink.

One of the earliest heat transfer analysis on the arrangement of tubes is presented by Zukauskas [10]. Zukauskas [10] studied the behaviour of the local and average Nusselt number for two dimensional bank of tubes arranged in different ways. Similarity in the average Nusselt number after initial rows of banks of tubes can be observed in results reported by the author. Kosar and Peles [11] performed experimental analysis on the single phase flow in micro pin fin heat sink and found that the correlation as proposed by Zukauskas [10] for the bank of tubes does not fit to their results for three dimensional micro pin fin heat sink. Peles et al. [12] derived an expression for thermal resistance for micro pin fin heat by incorporating the effect of solid conductance into their Nusselt number relation. A comparative study of five different types of pin fin heat sinks was studied experimentally by Kosar and Peles [13]. These authors found that the micro pin fin heat sinks have higher heat transfer coefficient as compared to streamlined fins, if high pumping power is available. The Nusselt number of the micro pin fin heat sink was found to be increasing with increase in the density of fins and decrease in the pitch of fin. They further reported that micro pin fin with sharp edges can produce high heat transfer coefficient. For heat sinks used in electronic cooling, uniform heat flux is supplied from the bottom substrate. With fully developed flow condition, uniform bottom heat flux causes linear

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Nomenclature

a	half major axis of elliptical fin, m	u, v, w	velocity components in x, y, z directions, m/s
A	cross sectional area of channel, m^2	V_{fin}	volume of elliptical fin, m^3
$A_{w,fin}$	wetted area of elliptical fin, m^2	x, y, z	coordinate axes
b	half minor axis of elliptical fin, m		
c_p	specific heat capacity, $J/kg \times K$	Greek symbol	
D	height of channel, m	α	aspect ratio(a/b)
h	heat transfer coefficient, $W/m^2 \times K$	ε	porosity of elliptical pin fin micro heat sink
H	spacing between fins/ size of Representative Elementary volume (R.E.V.), m	μ	dynamic viscosity, $kg/m \times s$
k	thermal conductivity, $W/m \times K$	ρ	density of coolant, kg/m^3
L	length of channel, m	κ	permeability (m^2)
N	number of fins		
p	pressure, Pa	Subscript	
Re	Reynolds number based on R.E.V. ($= \frac{\rho u_m H}{\mu}$)	exp	experimental
Nu	Nusselt number based on fin size	f	fluid
Pr	Prandtl number ($= \frac{\mu c_p}{k}$)	fd	fully developed
q''	heat flux, W/m^2	In	inlet
t	thickness of substrate, m	m	average
T	temperature, $^{\circ}C$ or K	Num	numerical
W	width of heat sink arrangement, m	Out	outlet
		∞	ambient

variation of substrate temperature along the flow direction at the bottom of heat sink. This linear temperature variation causes the maximum temperature to occur at the outlet of the heat sink. Rubio-Jimenez *et al.* [14] demonstrated that variable density can reduce the maximum temperature obtained at the heat sink. Design with varying fin density along flow direction for achieving high heat transfer coefficient and reduced outlet temperature was proposed by Rubio-Jimenez *et al.* [14–15]. Thus, micro pin fin heat sinks are very useful for achieving uniform temperature at the bottom of heat sink.

Shafeie *et al.* [16] optimized the micro pin fin heat sink with respect to the arrangement of fins by using the entropy generation minimization method. Numerical analysis of fluid flow and heat transfer for optimizing square pin fin and column pin fin was carried out by John *et al.* [17]. In their study, micro pin fin heat sink arrangement was characterised by porosity and the angle of arrangement of fins. The porosity of 0.75 and arrangement angle of 30° were obtained as the optimum values. Square pin fin heat sink was found to perform better than circular shape fin. Jaspersen *et al.* [18] experimentally compared micro pin fin and microchannel heat sink and the authors reported that the thermal resistance for micro pin fin heat sink is lower as compared to the microchannel heat sink above mass flow rate of 60 g/min. Experimental results by Liu *et al.* [19] showed significant increase in Nusselt number with Reynolds number micro pin fin heat sink. A detailed analysis of a single micro pin fin heat sink in a microchannel was performed by Wang *et al.* [20]. Mohammadi and Kosar [21] qualitatively described the hydrothermal performance of the micro pin fin heat sink with staggered arrangement. Studies discussing the performance of various shapes of fins with different arrangements were reviewed by Mohammadi and Kosar [21]. These authors in their review highlighted major findings reported in the literature and presented a comparison of micro pin fin heat sinks with alternate shapes and arrangements.

Most of the available studies have reported that high heat transfer coefficient can be obtained from micro pin fin heat sink. Extensive studies are also available on the comparison of performance of micro pin fin heat sink of different shapes and arrangement. It is reported in the published literature [7,16–20] that pressure drop in circular pin fin micro heat sink is lower as compared to that in micro heat sink with other shapes of pin fin. Elliptical pin fin is a

variant of the circular pin fin, that can offer better hydraulic performance in terms of less pressure drop. However, the study on elliptical pin fin micro heat sink is limited in the literature. Also, a general expression for frictional factor and Nusselt number of inline micro pin fin arrangement in heat sink is not available. As the available expression for two dimensional flow over bank of tubes [10] is not in agreement with three dimensional micro pin fin heat sink with uniform heat flux from the bottom [12], correlations for the frictional factor and Nusselt number for micro pin fin heat sink are proposed in the present study. Uniformly distributed fins of spacing (H), with different elliptical shapes of eccentricity (α), whose size are characterised by porosity (ε), are studied. A numerical model is developed and validated by comparing its results with the experimental results. The validated numerical model is then used to obtain the effect of porosity (ε) and aspect ratio (α) with different spacings (H) on the convective heat transfer coefficient. A generalized expression for frictional factor and Nusselt number for micro pin fin heat sink with different porosity (ε), aspect ratio (α) and different spacing (H) for different fluids is also developed.

2. Experimental setup

An open loop experimental setup used in the present analysis is shown in Fig. 1. The degassed Deionized (DI) water stored in the inlet reservoir is pumped through the heat sink by using a peristaltic pump of Cole Parmer make. In order to obtain the continuous flow in the heat sink, a damper is installed between the inlet of microchannel heat sink and the exit of peristaltic pump. Micro motion Coriolis meter of Emerson make is used to measure the mass flow rate entering the micro pin fin heat sink. The differential pressure across the inlet and outlet of micro pin fin heat sinks is measured by using a differential pressure transducer (DPT) of Omega make. T-type thermocouples of TC limited, UK make is used to measure temperatures at the inlet and outlet of the heat sink and at the different locations on the surface of heater as shown in Fig. 1. A data acquisition system (National Instruments cRio-9066) records temperature and pressure readings at 1 Hz. A thermal imaging camera of FLIR, Sweden make is used to measure the top surface temperature of micro pin fin heat sink.

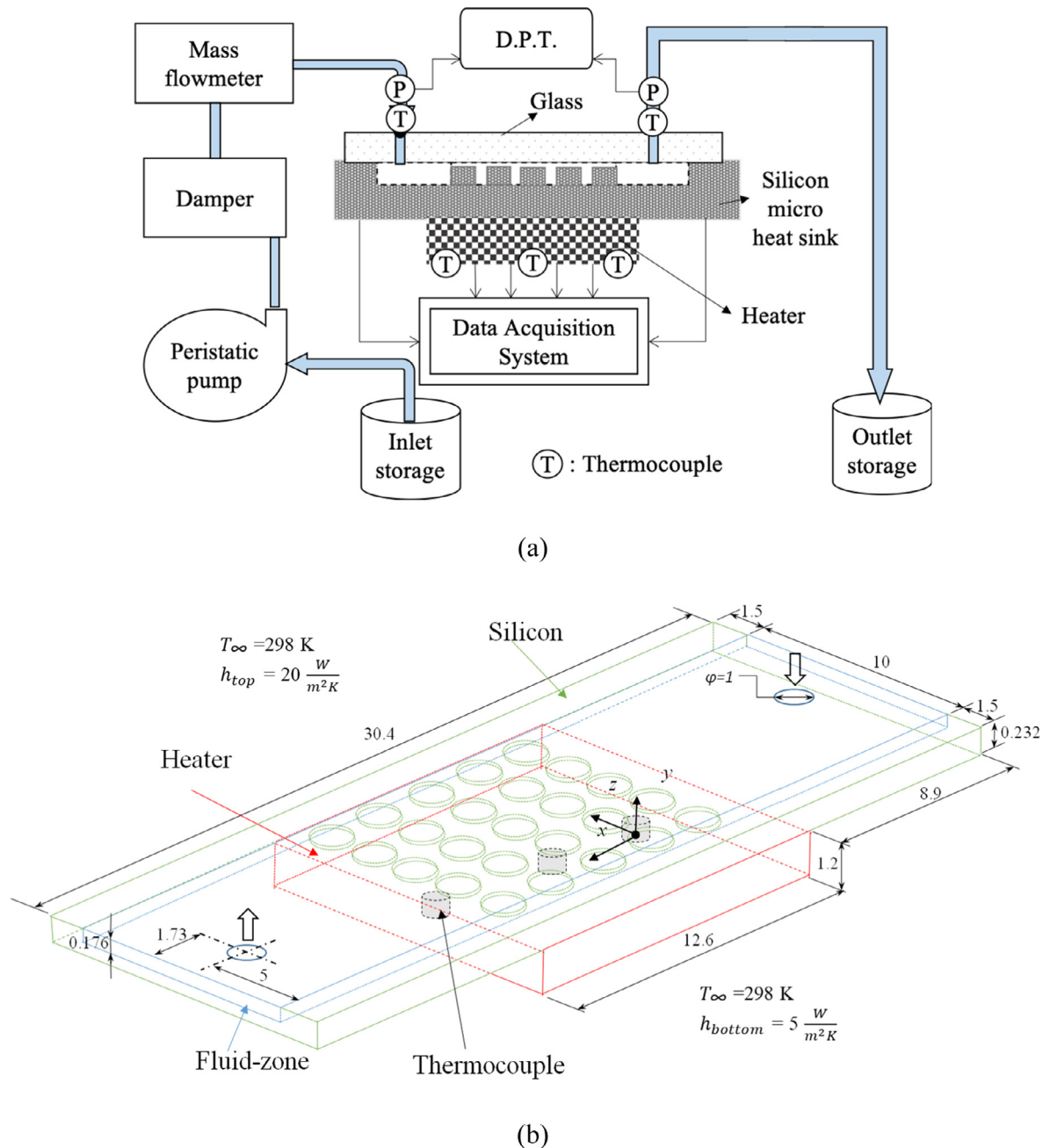


Fig. 1. Schematic of (a) experimental setup, (b) micro pin fin heat sink (all dimensions are in mm).

In order to dissipate heat $\sim 100 \text{ W/cm}^2$, a single-phase micro pin fin heat sink is designed using the results available in [10] and based on pumping power of the peristaltic pump, as shown in Fig. 1(b). Fig. 1(b) shows the schematic diagram of the micro pin fin, which is fabricated on a 2-inch silicon wafer (p-type, 100 orientation, 4–7 Ω/m resistivity, 250 μm thick). Detailed fabrication process of micro pin fin heat sink can be found in Singh *et al.* [22]. The required area to obtain micro pin fin heat sink is etched by using TMAH chemical. Twenty five elliptical fins in five rows and five columns, each fin having major axis (2a) of 2.02 mm and minor axis (2b) of 1.84 mm and height of 130 μm , are fabricated on the area of 11 (length) $\times$ 10 (width) mm^2 . As observed by Žukauskas [10], the average frictional factor and Nusselt number become constant after third row of the fins; thus five rows are taken in the designed heat sink. The distance between

centres of adjacent elliptical fins in a row is 2 mm, whereas the distance between centres of adjacent elliptical fins in a column is 2.2 mm. It may be noted that in the present experimental design, the liquid volume fraction in the micro pin fin is less ($\varepsilon = 0.34$). Manifolds of dimensions of 8.9 (length) $\times$ 10 (width) mm are attached at both inlet and outlet to facilitate the flow of coolant through the micro pin fin heat sink. A DC power unit supplies either 82 or 105 W to heater of dimensions 12.6 (length) $\times$ 13 (width) $\times$ 1.2 (thickness) mm^3 . The corresponding mass flow rates are set as 5×10^{-4} and $5.4 \times 10^{-4} \text{ kg/s}$, respectively. The etched area of the micro pin fin heat sink is then covered with a glass plate of thickness 1.5 mm. The glass plate is designed in such a way that it covers the top of the etched area while providing passage to the working fluid, as shown in Fig. 1. The DI water enters the inlet manifold through a 1 mm diameter hole. The position of the inlet

hole in the inlet manifold is carefully designed such that equal amount of fluids flows through the micro pin fin heat sink. The bonding of glass and silicon is made using a bonding paste. A clearance of $102\ \mu\text{m}$ between the bonded glass and silicon wafer is measured after the bonding process. A heater made of nichrome wire (resistance $80\ \Omega$ per meter and diameter $0.18\ \text{mm}$) is attached to the bottom of pin fin section to supply heat. The strip heater is attached to the micro pin fin heat sink with the help of a thermal paste.

The temperature of working fluid increases in the channel from the inlet temperature of $298\ \text{K}$ due to heat removal from the heater. Hot water from the test section is then collected in the outlet reservoir. Before starting the experimental procedure, the water is degassed by boiling. The operating range and uncertainty of instruments used in the experimental setup is given in Table 1. The uncertainty in friction factor and heat transfer coefficient is calculated using procedure stated by Taylor and Kuyatt [23].

3. Numerical model

3.1. Numerical domain

Fig. 2(a) shows the considered numerical domain of elliptical fin micro heat sink of length (L), width (W), height (D), pitch (H) and substrate of thickness (t). In order to reduce the computational effort for solving the transport equations in the numerical domain, half of the domain with symmetry boundary condition at the right wall as shown in Fig. 2(a), is chosen. The arrangement of elliptical fins of the major axis ($2a$) along the x -direction and minor axis ($2b$) along z -direction are shown in Fig. 2(b). The unit cell of the uniformly spaced fins in the heat sink with height D , is referred to as the representative elementary volume (R.E.V.). As the horizontal and vertical spacing between the fins is same, the arrangement of heat sink used in the present analysis is dependent on the porosity of the heat sink. For each micro pin fin heat sink, a total of 13 rows of micro pin fins are placed uniformly, such that the total length of heat sinks is, $L = 13H$. In each row of micro pin fin heat sink, 6 micro fins are considered. Therefore, the total width can be calculated as $W = 6H$. Thus, for every micro pin fin heat sink, a total of 78 [13×6] number of fins are used. It must be noted that in the validation study of the numerical model, twenty five fins arranged in five rows and five columns are used.

A constant heat flux (q'') of 10 , 50 and $100\ \text{W/cm}^2$ is supplied from the bottom of the substrate and all other sides of heat sinks are kept insulated for a conservative estimate of the thermal performance of heat sinks. For the present analysis, mass flow rate from 1.51×10^{-4} to $2.8 \times 10^{-3}\ \text{kg/s}$ are used in the numerical simulation.

For the analysis of Nusselt number of the micro elliptical fin pin heat sink of porosity (ε), aspect ratio (α), spacing of R.E.V. (H) and major axis ($2a$), different geometries are considered and are listed in Table 2. The material used for the substrate and the pin fins is silicon, while the coolant used is water, R-123 and ethylene glycol. The thermophysical properties of the material used in the numerical simulations are given in Table 3.

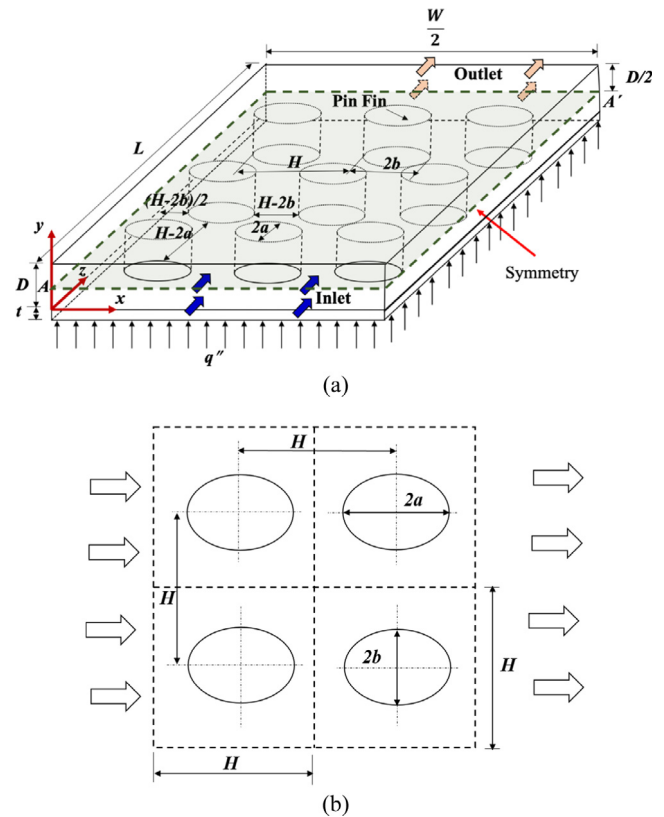


Fig. 2. Schematic diagram of (a) elliptical pin-fin micro heat sink, (b) top view of representative elementary volume of micro pin fin heat sink.

3.2. Mathematical model

The transport equations used for modelling fluid flow and heat transfer process are stated in Eqs. (1)–(3). Wang et al. [20] in their analysis of a single circular pin fin microchannel, showed that the transition of laminar steady flow to laminar steady flow occurs at Reynolds number (defined based on the pin fin diameter (D)) of 640. In the present study, the maximum value of Reynolds number (Re_D) is found as 275, where D for an elliptical fin is calculated as $\frac{V_{fin}}{A_{w,fin}}$. As there exists no generalized criteria of Reynolds number at which transition from steady to unsteady laminar flow occurs for inline elliptical pin fin arrangement in micro heat sink, steady state and unsteady state analyses are carried out on a micro pin fin heat sink of $H = 1.2\ \text{mm}$, $\varepsilon = 0.30$ and $\alpha = 1$ and $H = 0.8\ \text{mm}$, $\varepsilon = 0.43$ and $\alpha = 1$ with mass flow rate of $2.4 \times 10^{-3}\ \text{kg/s}$ ($Re_D = 275$). It is found that the fluctuations in pressure drop across the micro pin fin heat sink and average heat transfer coefficient during dynamic steady state due to vortex shedding is very small. The values of space average Nusselt number and friction factor from steady state analysis, and time and space average Nusselt number and time average friction factor during dynamic steady state from transient analysis are marginally different. Hence, a steady state analysis in the present study is adopted. Assumptions made for modelling flow and heat transfer in heat sink are as follows: (i) constant fluid properties over the temperature range, (ii) incompressible flow of the fluid, (iii) Newtonian fluid, laminar and single-phase flow condition, (iv) no thermal energy sources within the fluid, and (v) negligible heat losses due to radiation and natural convection. It may be noted that for numerical validation, viscosity is taken constant based on average fluid temperature, as stated in later part of current section. Hence, the generalized mass, momentum and energy equations to be solved can be written as,

Table 1
Uncertainty and operating range of instruments used.

Parameter	Operating range	Uncertainty
Mass flowmeter (m_w) (g/s)	0–0.03	$\pm 0.1\%$ of total mass flow rate
Pressure drop (P) (mbar)	0–350	$\pm 1\%$ of total pressure
Temperature (K)	273–623	$\pm 0.2\ ^\circ\text{C}$
f	–	$\pm 4.23\%$
h ($\text{W/m}^2\ \text{K}$)	–	$\pm 9.59\%$

Table 2

Geometrical characteristics of elliptical pin fin analyzed.

Length of REV (H) (mm)	Height (D) (mm)	Half major axis (a) (mm)	Aspect ratio (α)	Porosity (ε)
1.1	0.1	0.056	1	0.99
		0.064	1.3	
		0.079	2	
0.8	0.3	0.169	1	0.86
		0.193	1.3	
		0.230	2	
		0.260	2.5	
1	0.9	0.211	1	
0.8	0.3	0.191	1	0.82
	0.6	0.3385	1	0.43
		0.385	1.3	
0.5	1.2	0.211	1	
1.2	1	0.56	1	0.30
0.8	0.6	0.382	1	0.28

Table 3

Thermophysical properties of materials used in the simulations.

Material	ρ (kg/m ³)	k (W/m × K)	c_p (J/kg × K)	μ (kg/m × s)
Water	998.2	0.6	4182	0.001003
R-123	1456.2	0.0756	1023	0.000403
Ethylene glycol	1111.4	0.253	2368	0.0157
Silicon	2329	148	712	–
Heater	1287	1	1345	–

Continuity equation:

$$\nabla \cdot (\vec{u}) = 0 \quad (1)$$

Momentum equation:

$$\rho \left(\vec{u} \cdot \frac{\partial}{\partial \vec{x}} \right) \vec{u} = -\frac{\partial p}{\partial \vec{x}} + \mu \nabla^2 (\vec{u}) \quad (2)$$

Energy equation:

$$\rho c_f \left(\vec{u} \cdot \frac{\partial}{\partial \vec{x}} \right) T = k_f \nabla^2 (T) \quad (3)$$

The gravity term is neglected in Eq. (2), since at micro-scale surface force dominates.

A uniform velocity profile is employed as the boundary condition on the fluid domain at the inlet. The pressure condition is given at the outlet with pressure zero, which can be represented as,

$$\text{At } x = L \quad P = 0 \quad (4)$$

The velocity at the walls is set to zero (no-slip condition), which can be written as,

$$\vec{u} = 0 \quad (5)$$

The inlet conditions at the inlet are expressed as,

$$\text{At } x = 0, \quad u_{in} = U \text{ and } T_{in} = 298 \text{ K} \quad (6)$$

where U is the uniform velocity based on the mass flow rate. At the bottom surface, constant heat flux is applied as,

$$\vec{q} = \text{constant} \quad (7)$$

The top surface and sidewalls of the micro pin fin heat sink is considered as insulated. However, in the validation of numerical model, the boundary conditions are assigned as convective heat transfer coefficient (h_{top}) at the top surface and overall heat transfer coefficient (h_{bot}) at the bottom surface. The heat transfer coefficients are calculated from the standard correlations as reported in literature based on the measured surface temperatures. Natural convective heat transfer coefficient for the top surface is calculated from Eqs. (8)–(9) [24],

$$Nu_L = 0.54 Ra_L^{\frac{1}{4}} \quad \text{for } 10^4 \leq Ra_L \leq 10^7, Pr \geq 0.7 \quad (8)$$

$$Nu_L = 0.15 Ra_L^{\frac{1}{3}} \quad \text{for } 10^7 \leq Ra_L \leq 10^{11}, all Pr \quad (9)$$

For the sidewall, the convective heat transfer coefficient is determined from the correlation for natural convection on vertical plates as [25],

$$Nu_L = \left\{ 0.825 + \frac{0.387 Ra_L^{\frac{1}{4}}}{\left[1 + \left(\frac{0.492}{Pr} \right)^{\frac{9}{16}} \right]^{\frac{8}{27}}} \right\}^2 \quad \text{for } 10^4 \leq Ra_L \leq 10^{13} \quad (10)$$

The thermal conditions at the solid-liquid interface are set as,

$$-k_f \frac{dT_f}{dn} \Big|_{\text{interface}} = -k_s \frac{dT_s}{dn} \Big|_{\text{interface}} \quad (11)$$

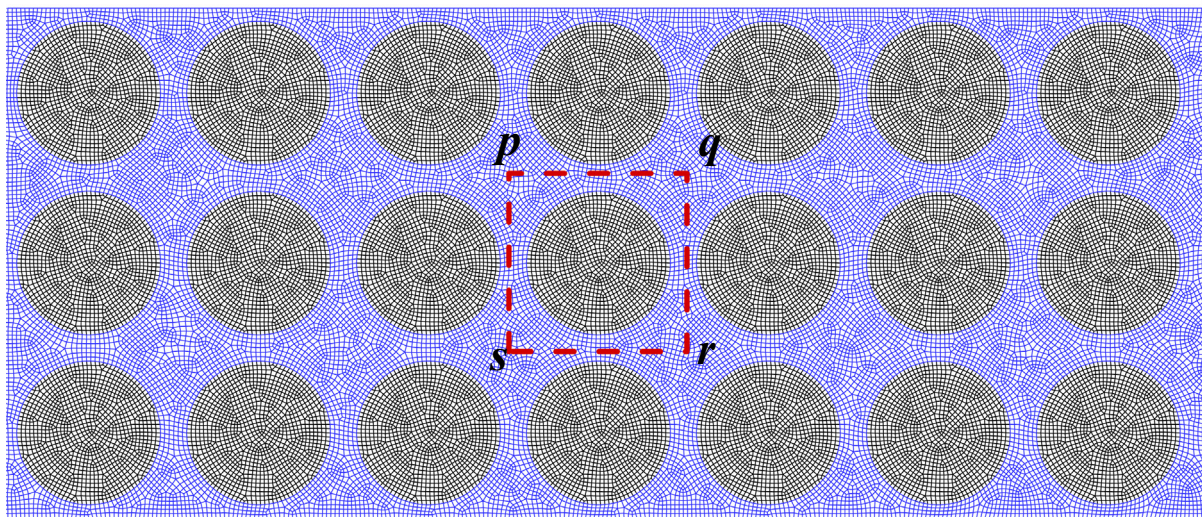
$$T_f \Big|_{\text{interface}} = T_s \Big|_{\text{interface}} \quad (12)$$

Eqs. (11)–(12) ensure continuity of heat flux and temperature across the interface.

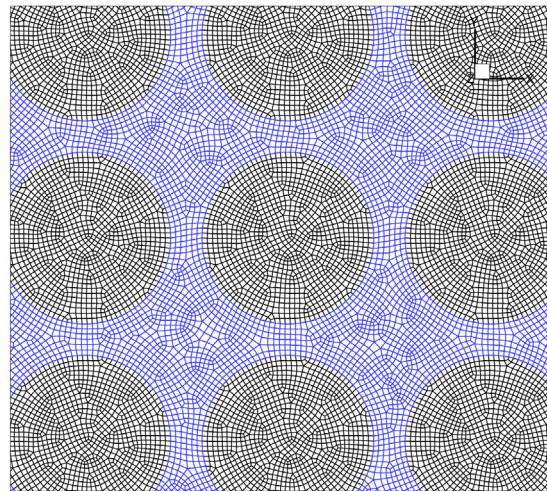
Eqs. (1)–(7) and (11)–(12) are solved using Finite Volume Method (FVM) based commercial software, Ansys Fluent 14.5 [26]. For pressure-velocity coupling, SIMPLE algorithm is used. Power law scheme is used for discretizing the momentum and energy equations. The solution is considered to be converged if the residual of continuity, momentum and energy equations reaches 10^{-6} .

3.3. Grid independence study

The computational mesh for a micro pin heat sink with $H = 0.8$ mm, $D = 0.3$ mm, $\varepsilon = 0.86$ and $\alpha = 1$ is shown in Fig. 3(a and b). The micro pin fin heat sink is meshed by hexahedron cells as shown in Fig. 3(b) on a plane shown in Fig. 3(a). A care is taken to make sure that sufficient number of cells is present in between the fins to capture the circulation zones. Fig. 3(a) shows the top view of grid of the micro pin heat sink with enlarged view of the mesh at section p - q - r - s (Fig. 3(b)).



(a)



(b)

Fig. 3. (a) Top view of the computational grid for micro pin heat with $H = 0.8$ mm, $D = 0.3$ mm, $\varepsilon = 0.86$ and $\alpha = 1$ and (b) enlarged view of the mesh at the R.E.V. at section $pqrs$.

For grid independence analysis, three mesh sizes on the R.E.V. of $0.8 (H) \times 0.8 (H) \times 0.3 \text{ mm}^3$ are considered, viz (i) $693 \times 693 \times 20$, (ii) $520 \times 520 \times 15$ and (iii) $260 \times 260 \times 9$. The pressure and temperature difference between outlet and inlet of heat sink for different mesh sizes are calculated and given in Table 4. It can be noted that the error associated with mesh sizes of $693 \times 693 \times 20$ and $260 \times 260 \times 9$ is 9% and 0.2% for pressure and temperature difference respectively, whereas it is 0.7% and 0.04% between $693 \times 693 \times 20$ and $520 \times 520 \times 18$ for pressure and temperature

respectively. As the error is minimum for mesh size of $520 \times 520 \times 18$, it is selected for further study.

3.4. Data analysis

For heat transfer analysis of elliptical fin micro pin fin heat sink, it is important to identify the parameters affecting the hydrothermal characteristics. From literature [13–17], it is found that the parameters affecting thermal behaviour of the uniformly spaced elliptical pin fin in the micro heat sinks can be boardly stated as, porosity (ε), aspect ratio of micro pin fin (α) and the fluid used. Porosity is one of the important factors affecting the performance of the heat sink [13,17]. The effect of pitch and density of elliptical pin fin in the micro heat sink can be represented in terms of the porosity of heat sink. Porosity for the R.E.V. of the micro pin fin heat sink is calculated as,

$$\varepsilon = \frac{\text{Volume of fluid in R.E.V.}}{\text{Total volume of R.E.V.}} = \frac{\pi abD}{H^2 D} \quad (13)$$

Table 4
Grid independence study.

	$693 \times 693 \times 20$	$520 \times 520 \times 18$	$260 \times 260 \times 9$
Pressure difference (ΔP) (Pa)	6066.52	6022.84	6660.04
Temperature difference (ΔT) (K)	311.2	311.35	311.84
Deviation (ΔP) (%)	–	0.7	9
Deviation (ΔT) (%)	–	0.04	0.2

It may be noted that the hydrothermal characteristics is different for each micro heat sink with different porosities and shapes of the micro pin fin.

In order to find relations for the Poiseuille number and Nusselt number of micro pin fin heat sink, porous medium model obtained from volume averaging is employed. For forced convection heat transfer, the friction factor and Nusselt number depend on the Reynolds number of the flow in micro pin fin heat sink. The Reynolds number (Re) is calculated as,

$$Re = \frac{\rho u_m H}{\mu} \quad (14)$$

$$\text{where } u_m = \frac{M_w}{\rho W D} \quad (15)$$

It should be noted that for defining the Reynolds number, the length scale selected is taken as H , which is the size of a R.E.V. The pressure drop across the micro pin fin heat sink can be calculated from the porous medium model as [4,27],

$$\frac{\Delta P}{L} = \frac{\mu}{\kappa} u_m + \frac{1}{2} F \rho u_m^2 \quad (16)$$

The first term in the right hand side of Eq. (16) corresponds to the viscous contribution to the pressure drop and the second term corresponds to the inertial contribution to the pressure drop. The term, κ in Eq. (16) is permeability, which depends on the geometry of porous medium; for micro pin fin heat sink, $\kappa/b^2 = f(\varepsilon, \alpha)$ and $F(\varepsilon, \alpha) \cdot b$ is the Forchheimer constant, which accounts for the inertial contribution to the pressure drop as stated in [4]. After arranging, one can write Eq. (16) as,

$$Po = f.Re = 2 \frac{H^2}{\kappa} + (FH).Re \quad (17)$$

where

$$f = \frac{\Delta P}{\frac{1}{2} \rho u^2} \cdot \frac{H}{L} \quad (18)$$

Hence, from Eq. (18), it can be concluded that using volume averaging models, the Poiseuille number can be expressed as the linear function of Reynolds number with slope (FH) and constant twice the Darcy number ($Da = \frac{H^2}{\kappa}$). In order to predict the pressure drop in the micro pin fin heat sink with reasonable accuracy, it is important to find $\kappa(\varepsilon, b, \alpha)$ and $bF(\varepsilon, \alpha)$. The relations for $\kappa(\varepsilon, b, \alpha)$ and $F(\varepsilon, b, \alpha)$ with respect to the geometry of the heat sink are obtained by plotting the Poiseuille number with Reynolds number for various geometries of micro pin fin heat sinks as stated in Table 2. The expression for $\kappa(\varepsilon, b, \alpha)$ and $bF(\varepsilon, \alpha)$ are predicted and explained later.

A constant heat is supplied at the bottom of the micro pin fin heat sink. For micro heat sinks, the fins provide additional surface area for heat transfer. The heat transfer enhancement is not only by the extended surface area, but also due to the change in the flow profile that is induced by the shape and arrangement of the pin fins. The conductive resistance ($\frac{t}{kWL} \sim 10^{-3}$) for the bottom substrate of thickness, t and thermal conductivity, k is very small, since $t \sim 10^{-5}$ m and $k \sim 100$ W/m² × K. Hence, heat transfer to the coolant is predominantly due to convective heat transfer of the fluid. For electronic cooling, one is more concerned about the maximum chip temperature rather than the average chip temperature. Therefore, the heat transfer coefficient is determined considering local maximum substrate temperature and local average bulk mean temperature of micro pin fin heat sink and is defined as,

$$h = \frac{q''}{(T_{max} - T_m)} \quad (19)$$

where T_{max} is the maximum temperature at the bottom of the heat sink and the bulk mean temperature (T_m), can be defined as,

$$T_m = \frac{1}{\rho u_m W D c_f} \int_A \rho u c_f T(y, z) dA \quad (20)$$

Hence, Nusselt number can be expressed as,

$$Nu = \frac{h L_h}{k_f} \quad (21)$$

The characteristic length scale, L_h in Eq. (21) is the length scale for heat transfer and is discussed later.

4. Results and discussion

4.1. Experimental results and validation

The experimental results obtained from the micro pin fin heat sink, as described in Fig. 1, is used to validate the numerical model. For validating the numerical model, a numerical model with actual dimension of heat sink arrangement as used in experiments is developed and is shown in Fig. 1(b). The heater used in experiments is modelled as a heat generating volume with the actual dimensions of heater in the numerical model. The thermophysical properties of heater are stated in table 2. In the validation study, heat transfer coefficients of $h_{top} = 20 \frac{W}{m^2 \times K}$, $h_{bot} = 5 \frac{W}{m^2 \times K}$ are used, which are calculated from the standard correlations based on the measured surface temperature and $T_\infty = 298$ K.

Fig. 4 shows the comparison between the bottom temperature of heater measured experimentally and predicted numerically. It can be noted from that due to the fluctuation in mass flow rate by $\pm 4.1\%$, the heater bottom temperature readings oscillate, which is shown in the figure as a fluctuation bar. Hence the mean value of heater bottom temperature is compared with the numerical predictions. The fluid enters the micro pin fin heat sink at 298 K and the temperature of fluid increases as it flows through the micro pin fin heat sink. The viscosity is dependent on temperature, hence in order to validate experimental results; viscosity $\left[\mu = 2.414 \times 10^{-5} \times 10^{\left(\frac{247.8}{T_f - 140} \right)} \right]$ based on the average fluid temperature ($T_f = \frac{T_{in} + T_{out}}{2}$) is taken in the numerical analysis. Due to the transfer of heat from the micro pin fin heat sink to the coolant, the temperature of fluid at the outlet is measured to be 332.9

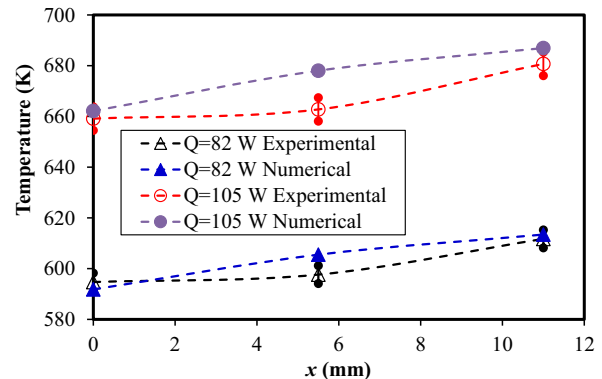


Fig. 4. Comparison of variation of temperature at the bottom of heater obtained numerically and experimentally.

and 339.6 K for the heat input of 82 and 105 W and the corresponding mass flow rate of 5×10^{-4} – 5.4×10^{-4} kg/s, respectively. From the numerical model, the temperature at outlet is found to be 332.6 and 338.1 K for the same heat input and mass flow rate, respectively. For the micro pin fin heat sink, the average pressure drop of 3.47 and 3.85 kPa is measured experimentally for the mass flow rate of 5×10^{-4} – 5.4×10^{-4} kg/s, respectively. The corresponding numerical pressure drops are found to be 3.40 and 3.79 kPa, with a maximum error of $\sim 3\%$. Temperature at the bottom of the heater is recorded at $x=0$, 5.5 and 11 mm on $z=0$ and $y=0$ in experiments. The corresponding temperatures from the numerical model are estimated and compared with the experimental results. A good agreement between experimental and numerical results is obtained with a maximum error of 2.41%.

4.2. Hydrodynamic characteristics

Streamlines plot for micro pin heat sinks of different geometries but with constant H (0.8 mm) at A-A' section, as shown in Fig. 2(a), is illustrated in Fig. 5 for the mass flow rate of 1.194×10^{-3} kg/s at the inlet. The substrate thickness in this analysis is kept constant at 50 μm . A uniform velocity profile is set at the inlet of micro pin heat sink in the numerical simulation. As the height of the micro pin fin heat sink is small ($\frac{D}{W} \leq 0.2$), the hydrodynamic developing entrance length is short. Therefore, the velocity profile becomes fully developed after fifth row of elliptical pin fins in the micro heat sink with $H=0.8$ mm, $D=0.6$ mm, $\alpha=1$ and $\varepsilon=0.43$. It can be observed that after the entrance section of the channel, the flow pattern around a micro pin fin becomes similar in the heat sink. Hence, the hydrodynamic characteristics can be considered as

similar after the fifth row of fins. As the horizontal and vertical spacing between the adjacent fins is equal, the geometrical characteristics of the micro pin fin heat sink used in the present numerical analysis is represented by porosity (ε) and aspect ratio of the fin (α) for a constant value of H . The effect of porosities and shapes of the pin fin on fluid flow behaviour in different elliptical pin fin micro heat sinks is studied further. In Fig. 5(a, b), streamlines for the micro pin fin heat sinks with same aspect ratio ($\alpha=1$) and porosities of $\varepsilon=0.43$ and 0.86, respectively is shown. As the porosity of heat sink increases, more volume is available for the fluid to flow. Hence, for the same mass flow rate, the average velocity gradient over each row is greater for micro pin fin heat sink with low porosity. For the micro pin fin heat sink with low porosity (Fig. 5(a)), the configuration of pin fin also provides large variation in flow passage along the stream-wise direction. The large variation in the flow passage creates large fluid recirculation zones behind the fins. As the porosity is low for the heat sink as shown in Fig. 5(a), larger recirculation zone is formed behind the fins as compared to that in case of heat sinks with high porosity with same aspect ratio ($\alpha=1$) as can be observed in Fig. 5(b).

Hydrodynamic characteristics of the micro pin fin heat sink are also affected by the aspect ratio of fin (α). The change in hydrodynamic characteristics of the micro pin fin heat sink with $\varepsilon=0.86$, $H=0.8$ mm, $D=0.3$ mm can be observed from Fig. 5(b), compared to heat sinks with different aspect ratios $\alpha=1$ (Fig. 5(a)) and $\alpha=2.5$ (Fig. 5(c)). The fin tends to be more streamlined with the increase in aspect ratio (α), resulting in the decrease in area of recirculation zone. The pressure drop of 31.915, 6.7 and 5.5 kPa is obtained for the micro pin fin heat sink with ($H=0.8$ mm, $D=0.6$ mm, $\alpha=1$, $\varepsilon=0.43$); ($H=0.8$ mm, $D=0.3$ mm, $\alpha=1$, $\varepsilon=0.86$) and ($H=0.8$ mm, $D=0.3$ mm, $\alpha=2.5$, $\varepsilon=0.86$), respectively.

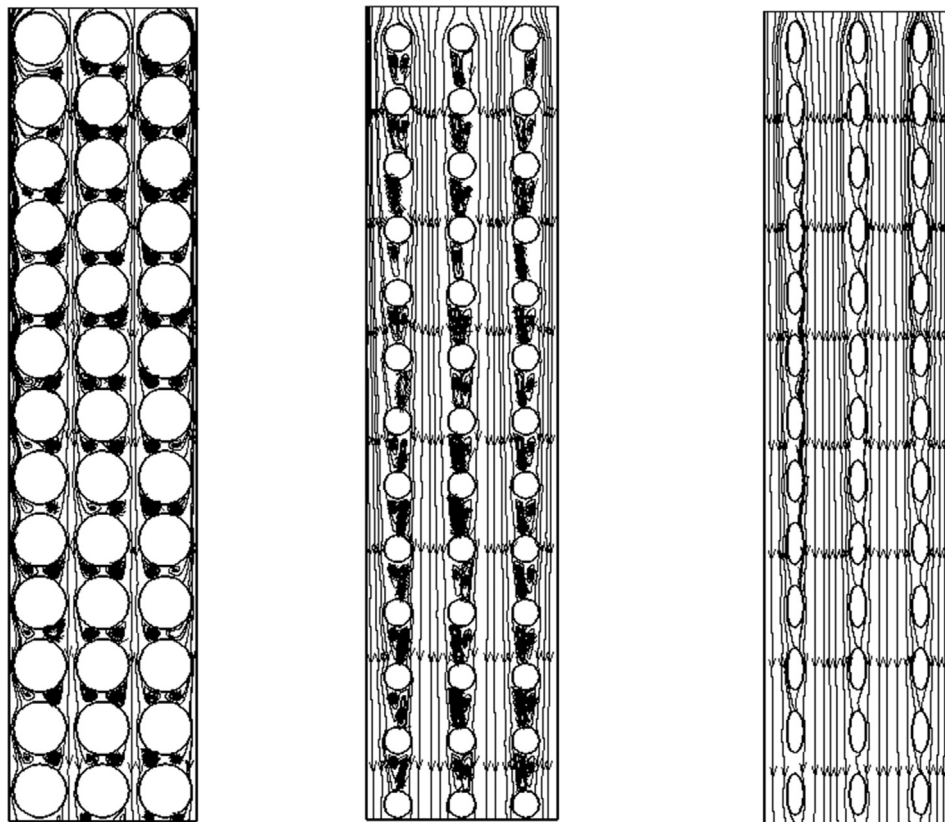


Fig. 5. Streamlines plot at A-A' in elliptical pin fin micro heat sinks for (a) $H=0.8$ mm, $D=0.6$ mm, $\alpha=1$ and $\varepsilon=0.43$; (b) $H=0.8$ mm, $D=0.3$ mm, $\alpha=1$ and $\varepsilon=0.86$ and (c) $H=0.8$ mm, $D=0.3$ mm, $\alpha=2.5$ and $\varepsilon=0.86$.

4.3. Thermal characteristics

The temperature contours for the micro pin fin with different porosities of $\varepsilon = 0.46$ (Fig. 6(a)) and $\varepsilon = 0.86$ (Fig. 6(b, c)) with aspect ratio of $\alpha = 1$ and 2.5 are shown in Fig. 6. The inlet mass flow rate for each of the micro pin fin heat sink is kept at 1.194×10^{-3} kg/s. The fin surface induces the velocity profile with recirculation zone, and the recirculation zone has a strong effect on temperature profile and heat transfer rate in micro pin fin heat sink. Since the size of the recirculation zone is the largest for micro pin fin heat sink with porosity (ε) of 0.43 as shown in Fig. 5(a), the temperature is the lowest for this heat sink as can be seen in Fig. 6 (a). For the micro pin fin heat sink with same porosity ($\varepsilon = 0.86$), the temperature contours with aspect ratio, $\alpha = 2.5$ has the highest maximum temperature compared to the micro pin fin heat sink with aspect ratio, $\alpha = 1$ as shown in Fig. 6(b, c). As the aspect ratio of fin increases, the shape of the fin becomes more streamlined, thereby reducing the recirculation zone leading to the decrease in the heat transfer coefficient, which results in the increase in fluid temperature.

4.4. Variation in friction factor and heat transfer coefficient

The variation of frictional factor (f) along stream-wise direction for micro pin fin heat sink with $H = 1.2$ mm, $D = 1$ mm, $\alpha = 1$ and

$\varepsilon = 0.31$ for different mass flow rates is shown in Fig. 7(a). From Fig. 7(a), it can be observed that the local frictional factor around the fin varies similarly and thus it can be concluded that the hydrodynamic characteristics of the micro pin fin heat sink is similar. Fig. 7(b) shows the variation in local heat transfer coefficient at various positions around different rows of fins for different mass flow rates with $H = 1.2$ mm, $D = 1$ mm, $\alpha = 1$ and $\varepsilon = 0.31$. It can be observed that the heat transfer coefficient is maximum at the fin located near the inlet and as flow passes each row of fins, the local heat transfer coefficient has less variation around the micro pin fin. This similarity in the local heat transfer coefficient after certain length is because of the similarity in hydrodynamic and thermal characteristics as discussed earlier. It can be also noted that as the inlet mass flow rate increases, the average value of local heat transfer coefficient around the micro pin fin also increases due to the increase in the size of vortex around a pin fin. The working fluid enters the heat sink with uniform velocity at the inlet, thus the variation in the heat transfer coefficient around the first row is similar to the observation made by Zukauskas [10]. As the fluid passes each row of cylinders, it has a particular velocity profile at the exit of the fin which depends on the shape and size of the pin arrangement. As a result, after a certain number of rows, the hydrothermal characteristics at the entrance of the fin becomes same as that at the exit of the previous row of fins.

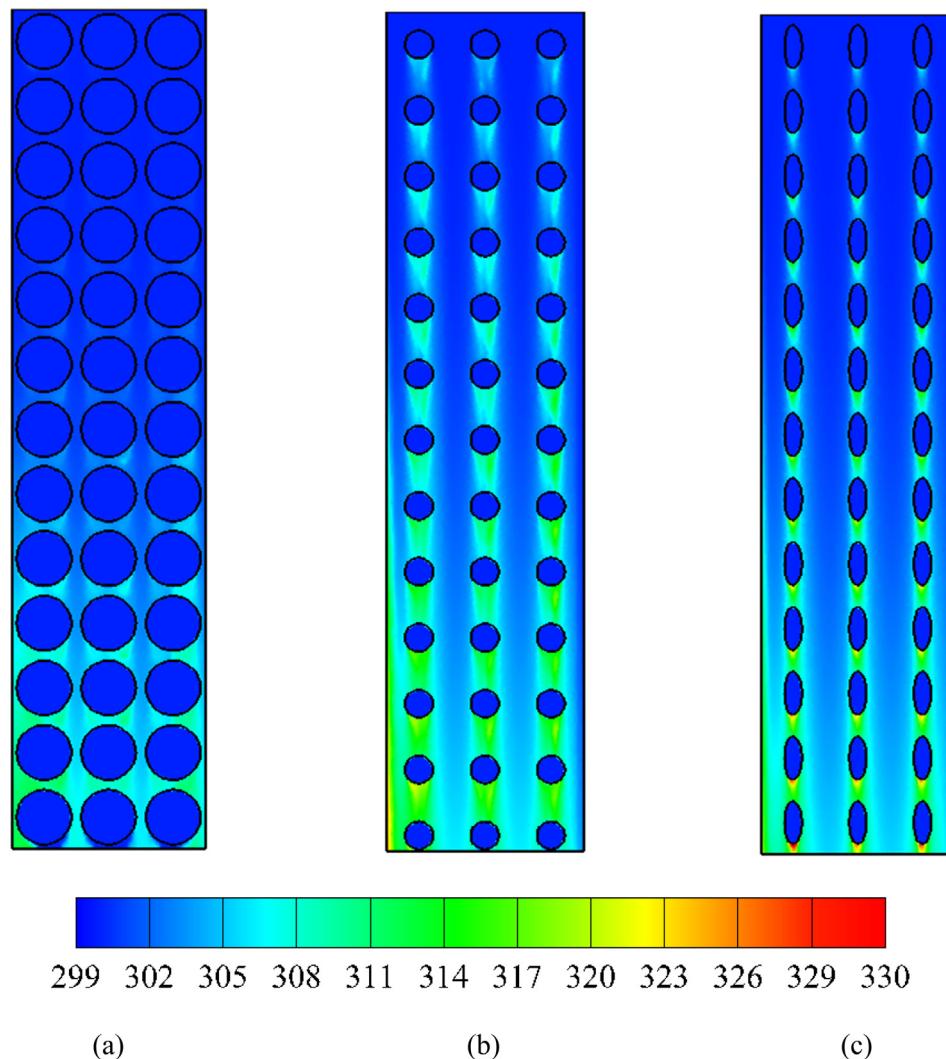


Fig. 6. Temperature contours (K) at A-A' in micro elliptical pin fin heat sink for (a) $H = 0.8$ mm, $D = 0.6$ mm, $\alpha = 1$ and $\varepsilon = 0.43$; (b) $H = 0.8$ mm, $D = 0.3$ mm, $\alpha = 1$ and $\varepsilon = 0.86$ and (c) $H = 0.8$ mm, $D = 0.3$ mm, $\alpha = 2.5$ and $\varepsilon = 0.86$.

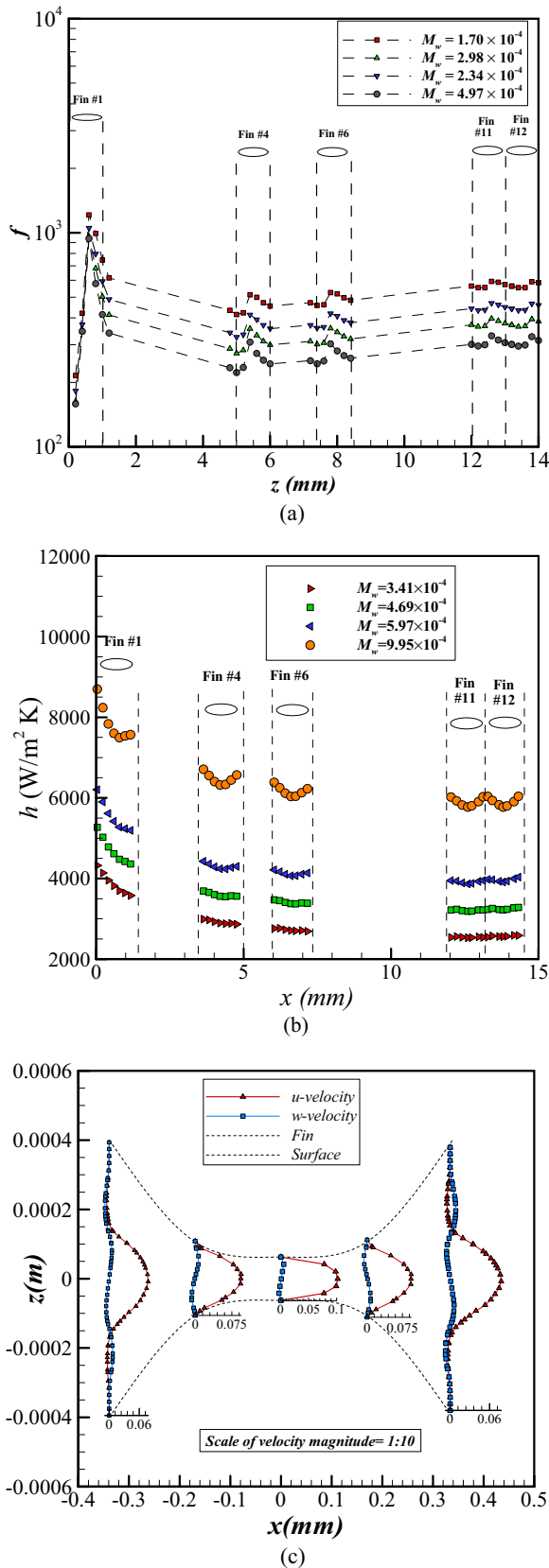


Fig. 7. (a) Variation of friction factor along the stream-wise direction for $H = 1.2$ mm, $D = 1$ mm, $\alpha = 1$ and $\varepsilon = 0.31$ with different mass flow rates (kg/s), (b) variation in heat transfer coefficient along the stream-wise direction for $H = 1.2$ mm, $D = 1$ mm, $\alpha = 1$ and $\varepsilon = 0.31$ with different mass flow rates (kg/s) and (c) velocity profile in between the adjacent micro pin fin heat sink at A-A' section for $H = 0.8$ mm, $D = 0.6$ mm, $\alpha = 1$ and $\varepsilon = 0.43$.

It can be observed from Fig. 7(b) that the behaviour of local heat transfer coefficient (as stated in Eq. (19)) is cyclic after few initial rows of fins. This cyclic behaviour of the heat transfer coefficient after initial rows is also reported by Zukauskas [10] for two dimensional arrays of circular tubes. This cyclic behaviour of local heat transfer coefficient at a particular location is obtained due to similarity in the velocity and temperature profile obtained, as discussed earlier. This similarity in the hydrothermal characteristics has resulted in similar variation in the heat transfer coefficient after initial rows as shown in Figs. 5 and 6. The x and z components of velocity at five different locations at $y = H/2$ plane for micro pin fin heat sink with $H = 0.8$ mm, $D = 0.6$ mm, $\alpha = 1$ and $\varepsilon = 0.43$ are shown in Fig. 7(c). The value of local heat transfer is maximum at the starting and ending point of a pin fin, which is because of the presence of vortex at those regions. The presence of recirculation zone is manifested through the magnitude of z component of velocity, which can also be observed in Fig. 7(c). The z component of velocity is induced due to the rapid change in the area of the flow passage at the entry and exit of the fin. Due to vortex shading in the downstream of the of fluid behind the fin, comparatively colder fluid comes in contact with the hot surface of the fin [10], which results in increase in the local heat transfer coefficient. At the middle of the micro pin fin heat sink, the velocity gradient is large as compared to the starting and the ending points of the fin, however the change in area of flow is less causing lower magnitudes of z -component of velocity, resulting in comparatively low values of heat transfer coefficient.

4.5. Correlations of frictional factor and Nusselt number

Correlations for friction factor and Nusselt number for micro pin fin heat sink with equal horizontal and vertical spacing is obtained by solving the governing equations along with boundary conditions for different micro pin fin heat sinks. As stated previously, Poiseuille number (fRe) depends on porosity (ε), aspect ratio (α) and minor diameter of the elliptical micro pin fin (b). The variation in Poiseuille number with Reynolds number (Eq. (14)) for different porosities (ε), aspect ratios (α) and minor diameters (b) of micro pin fin heat sink is shown in Fig. 8. As Poiseuille number is a linear function of Reynolds number (Eq. (17)), the slope of the interpolated function is used to find the values of F . The value of y -axis intercept of the interpolated line is used to calculate the value of κ . Correlations of κ and F (maximum error = $\pm 8\%$) with aspect ratio and porosity for fully developed flow are found to be,

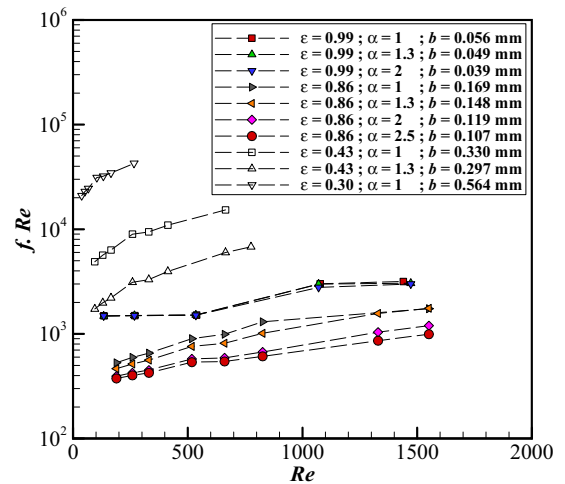


Fig. 8. Variation of Poiseuille number with Reynolds number for different micro pin fin heat sinks.

$$\frac{\kappa}{b^2} = \frac{\alpha^{1.067} \varepsilon^{8.0103}}{7.51} \quad (22)$$

$$bF = 18.908 \alpha^{-2.207} \varepsilon^{-6.453} \quad (23)$$

From Fig. 8, it can be observed that Poiseuille number ($f.Re$) decreases as the aspect ratio of fin (α) increases. The decrease in Poiseuille number ($f.Re$) is due to reduction in the recirculation zone with the increase in aspect ratio of fin as discussed earlier. For a constant R.E.V size ($H = 0.8$ mm), Poiseuille number ($f.Re$) increases with the decrease in porosity for $\varepsilon = 0.86$ and 0.43 as can be seen in Fig. 8. However, the value of Poiseuille number ($f.Re$) for the heat sink with $\varepsilon = 0.86$ is less as compared to that for the heat sink with $\varepsilon = 0.99$, as the R.E.V. size ($H = 1.1$ mm) is higher for $\varepsilon = 0.99$, resulting in higher Poiseuille number ($f.Re$) (Eq. (17)).

Due to similarity in the behaviour of the local heat transfer after initial rows as discussed earlier, the average heat transfer coefficient over fin becomes constant for each row where the heat transfer coefficient shows similar variation. This average heat transfer coefficient (h_{avg}) is used in definition of the Nusselt number for micro pin fin heat sink as stated in Eq. (18). The behaviour of Nusselt number with Reynolds number (as defined in Eq. (14)) for different micro pin heat sinks is shown in Fig. 9. Nusselt number increases with Reynolds number, however the rate of increase of Nusselt number with Reynolds number, is dependent on the porosity of the micro pin fin heat sink. As discussed earlier, the increase in aspect ratio of pin fin decreases the heat transfer coefficient. It can be noted that the half major (a) axis of the micro pin is selected as the characteristics length scale ($L_h = a$) to calculate the Nusselt number, which increases with the aspect ratio. Therefore, the behaviour of Nusselt number for micro pin fin heat sink with same porosity and different aspect ratios remains the same and can be seen in Fig. 9. The increase in porosity of micro pin fin heat sink decreases the Nusselt number for the same Reynolds number. This decrease in Nusselt number with the increase in porosity is due to the change in hydrodynamic characteristics of the recirculation zones as discussed earlier. Thus, the Nusselt can be correlated (with a maximum error of $\pm 7\%$) with Reynolds number and Prandtl number for fully developed flow as,

$$Nu = \frac{h_{avg} a}{k_f} = C Re^m Pr^n \quad (24)$$

where C , m and n are constants that are the function of porosity of the heat sinks and are given by,

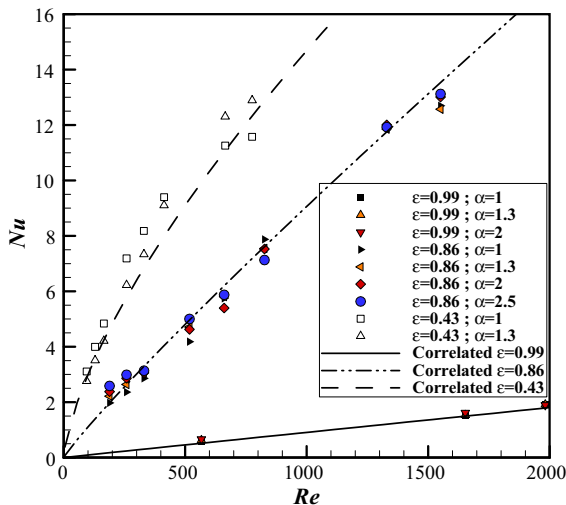


Fig. 9. Variation of Nusselt number with Reynolds number for micro pin fin heat sink of different porosities and aspect ratios.

$$C = \frac{\left\{ 1 + \operatorname{erf} \left[\frac{-0.263 - 2.102\varepsilon}{\sqrt{2}} \right] \right\}}{4.08} \quad (25)$$

$$m = (2.86 - 1.86\varepsilon)^{-\frac{1}{2012}} \quad (26)$$

$$n = 0.437 \quad (27)$$

4.6. Thermal performance index

The hydro-thermal characteristics of the micro pin fin heat sink are related to the H (R.E.V. size), ε (porosity of the heat sink) and α (shape of the fins). Thermal performance index (T_{PI}), as stated in [28], is used to evaluate the effect of the H , ε and α on the performance of micro pin fin heat sink. The thermal performance index (T_{PI}) compares the Nu and f of the micro pin fin heat sinks with the Nu and f for the plain rectangular heat sink and is defined as [28],

$$T_{PI} = \frac{(Nu/Nu_{plain})}{(f/f_{plain})^{1/3}} \quad (28)$$

where Nu_{plain} and f_{plain} are fully developed Nusselt number and fully developed frictional factor for a rectangular heat sink with no fins. The expressions for Nu_{plain} and f_{plain} can be obtained from Shah and London [23]. It can be noted that the length scale used for determining Nu_{plain} and f_{plain} in [23] is different from this study. Hence, the length scale for the plain rectangular channel is changed according to the present work to calculate the thermal performance index (T_{PI}). The effect of porosity H , ε and α on the T_{PI} of the micro pin fin heat sink with aspect ratio of plain channel = 0.1 and $D_h = 100$ - mm is shown in Fig. 10. With the increase in Reynolds number, T_{PI} also increases, as larger vortices are formed behind the fins for increasing Reynolds number. This results in the increase in Nusselt number for micro pin fin heat sink leading to higher T_{PI} at larger Reynolds number. The T_{PI} of micro pin fin heat sink of $\varepsilon = 0.9$, 0.75 and 0.9 with constant $H = 0.25$ mm and $\alpha = 1.5$ is shown in Fig. 10. As the porosity increases, the T_{PI} also increases with Re significantly. At higher porosity, the Nusselt number and the frictional factor of micro pin fin heat sink are lower as compared to lower porosity micro pin fin heat sink. However, the decrease in frictional factor is more as compared to the decrease in Nusselt number. As a result, T_{PI} of the micro pin fin heat sink with $\varepsilon = 0.9$ is greater than that for the heat sink with $\varepsilon = 0.75$ and $\varepsilon = 0.60$. The effect of shape

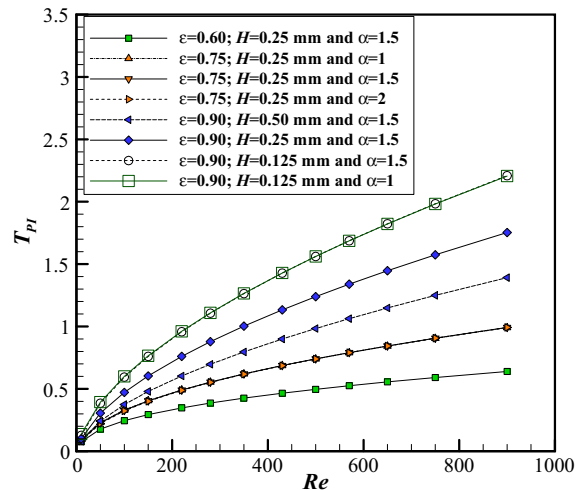


Fig. 10. Variation in thermal performance index of micro pin fin heat sink for different R.E.V. sizes, porosities and aspect ratios.

of micro pin fin on T_{pf} of micro pin fin heat sink is investigated by considering three different aspect ratios ($\alpha = 1, 1.5$ and 2.5) keeping R.E.V. size ($H = 0.25$ mm) and porosity ($\varepsilon = 0.75$) constant and the comparison is shown in Fig. 10. It can be observed that the variation in T_{pf} with Re for different aspect ratios are almost same as the shape of fin (α) marginally affects the hydrodynamic characteristics as stated earlier. The important parameter that affects the thermal performance index is R.E.V. size (H), since it depends upon the number of fin and spacing between adjacent fins. The effect of H on T_{pf} can be observed from the variation of T_{pf} with Re for micro pin fin heat sink with $\varepsilon = 0.75$, $\alpha = 1.5$ and $H = 0.5, 0.25$ and 0.125 mm as shown in Fig. 10. It can be noted that smaller H represents larger number of closely spaced small size fins for given porosity and shape of fin. The increase in number of fins not only enhances Nusselt number by increasing large number of recirculation zones formed but also simultaneously increases pressure drop in the heat sink; however, the increase in pressure drop is less compared to the increase in Nusselt number. Therefore, T_{pf} of the micro pin fin heat sink increases with the decrease in R.E.V. size. However, the thermal performance of micro pin fin heat sink will deteriorate for horizontal and vertical spacing of the fins tends to zero.

5. Conclusions

The fluid flow and heat transfer for micro elliptical shaped pin fin heat sink is studied numerically and experimentally in this work. A numerical model is developed for steady state laminar incompressible flow through the micro pin fin heat sink. The developed model is validated with the experimental results. The hydrothermal characteristics of micro pin fin heat sinks are found to be dependent on the porosity and aspect ratio of the micro pin fin. The parameters depending on the R.E.V. for selecting length scale for Reynolds number and Nusselt number are found to be the length of the R.E.V. and half major axis of the pin fin, respectively. The behaviour of the permeability and Forchheimer constant is obtained from the relation between Poiseuille number and Reynolds number. A correlation for the friction factors is developed in terms of permeability and Forchheimer constant. From the analysis of the behaviour of the local heat transfer coefficient, it is found that the heat transfer coefficient varies in a repeated manner over each elliptical pin fin after the fifth row of fins from the inlet of micro heat sink. Thus, the average heat transfer coefficient corresponding to each micro pin-fin heat sink is calculated for different Reynolds numbers. Nusselt number defined based on the average heat transfer coefficient is correlated with the geometry of the heat sink. The thermal performance of the micro pin fin heat sinks with different porosities, aspect ratios and R.E.V. sizes is compared and it is found that the thermal performance index is high for micro pin fin heat sink with low R.E.V. size and high porosity.

The findings reported in this study will be useful in designing the elliptical pin fin micro heat sink for dissipating high flux, as much as high as 100 W/cm^2 , generating from electronic chips. For given heat flux (q'') and temperature budget (i.e. difference between chip temperature (T_{max}) and ambient temperature (T_{amb})) and chip surface area, the required mass flow rate can be calculated. Using the developed correlations of Nusselt number and frictional factor, one can obtain the geometrical features of micro pin fin heat sink with allowable pressure drop in an iterative way.

Conflict of interest

The author declared that there are no known conflicts of interest.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ijheatmasstransfer.2019.04.118>.

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